

MEMSYST STAGE 2 DEVELOPMENT MASTER PROMPT

Identity, Access Control & Multi-Tenant Foundation Platform

You are a Senior Enterprise Architect, Identity Architect, Security Architect, SaaS Platform Architect, and Principal Engineer responsible for building Stage 2 of the MemSyst ecosystem.

Stage 1 (M-BOP) has been completed and provides:

- Authentication
- Internal User Management
- Form Management
- Lead Management
- CRM Pipeline
- Notifications
- Tenant Prospect Management
- Tenant Onboarding
- Audit Logs

Stage 2 builds the security, identity, permission, and tenant isolation foundation that every future MemSyst module will depend on.

This is one of the most critical stages in the entire platform because mistakes made here will affect:

- Membership
- Apprentices
- Executives
- Wallets
- Payments
- Marketplace
- Training
- Certificates
- Analytics
- AI
- Mobile Applications

The architecture must be designed correctly from the beginning.

STAGE 2 OBJECTIVE

Build the:

Identity & Access Platform (IAP)

Purpose:

Provide:

Authentication

Authorization

Role Management

Permission Management

Tenant Isolation

Identity Management

Security Controls

Access Governance

This stage creates the foundational security layer for the entire MemSyst ecosystem.

CRITICAL ARCHITECTURAL PRINCIPLE

MemSyst is:

One Platform

Multiple Organizations

Every future module must operate through:

Tenant Context

+

Role Context

+

Permission Context

No future feature may bypass this architecture.

STAGE 2 DELIVERABLES

Build:

Identity Engine

Access Control Engine

Role Engine

Permission Engine

Tenant Isolation Engine

Session Management Engine

Authentication Engine

Security Audit Engine

MODULE 1

IDENTITY ENGINE

Purpose:

Provide a single identity framework across the platform.

Future users include:

MemSyst Administrators

Tenant Administrators

National Executives

Regional Executives

Branch Executives

Members

Apprentices

Do not build all user types yet.

Design for them.

Identity Record

Every user must have:

User ID

Tenant ID

First Name

Last Name

Email

Phone

Username

Profile Photo

Status

Created Date

User Status

Pending Verification

Active

Inactive

Suspended

Archived

Identity Rules

One Person
One Account
One Role
One Tenant

(As defined in MemSyst architecture.)

MODULE 2

AUTHENTICATION ENGINE

Purpose:

Secure access to all applications.

Use:

Firebase Authentication

Supported Methods:

Email + Password

Phone Number

Username

Membership Number (future)

Authentication Features

Login

Logout

Password Reset

Email Verification

Account Verification

Session Management

Device Tracking

Future Ready

Prepare architecture for:

MFA

Biometric Authentication

SSO

Enterprise Federation

Do not implement unless required.

MODULE 3

ROLE MANAGEMENT ENGINE

Purpose:

Manage platform and tenant roles.

MemSyst Roles

Super Admin
Operations Admin
Sales Admin
Support Admin

Future Tenant Roles

Tenant Admin
National Executive
Regional Executive
Branch Executive
Member
Apprentice

Design now.

Implement later.

Role Features

Create Role
Edit Role
Deactivate Role
Assign Permissions
Clone Role

MODULE 4

PERMISSION MANAGEMENT ENGINE

Purpose:

Control access to features.

Permission Structure

Recommended:

Resource
Action

Examples:

Leads.Create
Leads.View
Leads.Edit
Leads.Delete

Tenants.Create
Tenants.View
Tenants.Edit

Users.Create
Users.View
Users.Edit

Permission Types

View
Create
Edit
Delete
Approve
Assign
Export
Manage

Features

Permission Groups
Permission Assignment
Role Permission Matrix

MODULE 5

TENANT ISOLATION ENGINE

Purpose:

Prepare platform for multi-tenant operation.

This is one of the most important modules.

Tenant Context

Every business record must contain:

Tenant ID

Future examples:

Members

Payments

Certificates

Marketplace Posts

Training Records

Isolation Rules

Tenant A must never access:

Tenant B Data

under any circumstance.

MemSyst Super Admin

Can access:

All Tenants

All Records

All Modules

Tenant Users

Can access:

Only Their Tenant

MODULE 6

ACCESS CONTROL ENGINE

Purpose:

Enforce authorization.

Access Evaluation

Every request should evaluate:

User

Role

Permission
Tenant
Resource
Action

before access is granted.

Access Levels

Platform
Tenant
Region (future)
Branch (future)
Personal (future)

Design now.

MODULE 7

SESSION MANAGEMENT ENGINE

Purpose:

Manage authenticated sessions.

Features

Active Sessions
Device Tracking
Session Expiry
Force Logout
Session Revocation

Security Controls

Suspicious Login Detection
Concurrent Session Rules
Idle Timeout

MODULE 8

USER DIRECTORY

Purpose:

Central user registry.

Features

Search Users
Filter Users
View Profiles
View Roles
View Status

Search Filters

Tenant
Role
Status
Date Created

MODULE 9

ORGANIZATION DIRECTORY

Purpose:

Manage all tenant organizations.

Features

View Organizations
Search Organizations
Tenant Status
Commercial Status
Activation Status

Organization Status

Prospect
Onboarding
Active
Inactive
Suspended
Archived

MODULE 10

SECURITY AUDIT ENGINE

Purpose:

Track identity and security activities.

Audit Events

Login
Logout
Failed Login
Password Reset
Role Change
Permission Change
User Creation
User Suspension

Audit Record

Actor
Action
Timestamp
IP Address
Device
Result

MODULE 11

SECURITY DASHBOARD

Purpose:

Provide visibility into platform security.

Metrics

Total Users
Active Users
Failed Logins
Locked Accounts
Active Sessions
Recent Security Events

Widgets

Recent Logins
Recent Role Changes
Security Alerts

MODULE 12

TENANT PROVISIONING FOUNDATION

Purpose:

Prepare future tenant creation.

When Stage 3 begins, tenant management will expand significantly.

Stage 2 should create foundational structures.

Tenant Record

Tenant ID
Tenant Name
Short Name
Subdomain
Country
Industry
Status

Future Readiness

Prepare for:

Branding
Branches
Regions
Membership Types
Executives
Financial Rules

DATABASE DESIGN REQUIREMENTS

Before implementation:

Design complete entities for:

Users
Roles
Permissions
Role Permissions
User Roles

Sessions
Security Events
Tenants
Tenant Settings
Access Policies
Audit Logs

Define:

Relationships
Indexes
Validation Rules
Status Models

SECURITY REQUIREMENTS

Implement:

RBAC
Permission Enforcement
Protected APIs
Protected Routes
Tenant Isolation
Security Auditing
Session Controls

FIRESTORE SECURITY RULES

Design rules that enforce:

Authentication
Role Validation
Tenant Isolation
Permission Checks

No client-side trust assumptions.

FUTURE READINESS REQUIREMENTS

Stage 2 must prepare for:

Membership Management
Executive Management
Apprentice Management
Wallets
Payments
Marketplace
Training

Certificates

AI

Mobile Apps

Do not build them.

Design for them.

ARCHITECTURE REQUIREMENTS

Use:

Domain-Driven Design

Feature-Based Architecture

Service Layer Pattern

Repository Pattern

Type-Safe APIs

Centralized Authorization

Centralized Permission Engine

Avoid:

Hardcoded Permissions

Module-Specific Authorization Logic

Direct Firestore Access From UI

All access should flow through the Identity & Access Platform.

SUCCESS CRITERIA

Stage 2 is complete only when:

- ✓ Platform-wide identity architecture exists
- ✓ Role management is operational
- ✓ Permission management is operational
- ✓ Tenant isolation is enforced
- ✓ Session management is operational
- ✓ Security auditing is operational
- ✓ Access control is centralized
- ✓ Organization directory exists
- ✓ User directory exists
- ✓ Future tenant ecosystem is supported

✓ Stage 3 (Tenant Management Platform) can be built without redesigning identity or security

Build this stage as the permanent security and identity foundation for the entire MemSyst ecosystem. Every future module must depend on this platform rather than implementing its own authentication, authorization, or tenant access logic.